INDEX

INSTITUTE

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION
MISSION

DEPARTMENT

VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

PROGRAM EDUCATIONAL OBJECTIVES, PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations**.** **[K3]**
6. **The Engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication**: Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance**: Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

Grams: "TECHNOLOGY"
Email: dapjntuk@gmail.com

Phone: 0884-2300991
Mobile: +9963993504

Directorate of Academic & Planning
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA
KAKINADA-533003, Andhra Pradesh, INDIA
(Established by AP Government Act No. 30 of 2008)

Lr. No. JNTUK/DAP/AC/B. Tech/IV Year/2019-20 *Date: 30-05-2019*

Dr. A. Mallikarjuna Prasad
M.E, Ph.D.,
Director, Academic Planning

To
All the Principals of Affiliated Colleges,
JNTUK, Kakinada

ACADEMIC CALENDAR FOR B.TECH IV YEAR (2016 BATCH)

I SEMESTER			
Description	**From**	**To**	**Weeks**
Commencement of Class Work	**10.06.2019**		
I Unit of Instructions	10.06.2019	03.08.2019	8W
I Mid Examinations	05.08.2019	10.08.2019	1W
II Unit of Instructions	12.08.2019	05.10.2019	8W
II Mid Examinations	07.10.2019	12.10.2019	1W
Preparation & Practicals	14.10.2019	19.10.2019	1W
End Examinations	21.10.2019	02.11.2019	2W
Commencement of II Semester Class Work	**18.11.2019**		
II SEMESTER			
I Unit of Instructions	**18.11.2019**	11.01.2020	8W
I Mid Examinations	13.01.2020	23.01.2020	1W
II Unit of Instructions	24.01.2020	21.03.2020	8W
II Mid Examinations	23.03.2020	28-03-2020	1W
Preparation	30.03.2020	04.04.2020	1W
End Examinations	06.04.2020	18.04.2020	2W

A.m.prasad
Director Academic Planning

Copy to the Secretary to the Hon'ble Vice Chancellor, JNTUK.
Copy to PA to the Rector, JNTUK.
Copy to PA to the Registrar, JNTUK.
Copy to PA to the Director of Evaluation, JNTUK.

CO-CURRICULAR AND EXTRA CURRICULAR ACTIVITIES

05/06/2019 – World Environment Day	15/09/2019 – Engineers Day		21/03/2019 – International Forest Day
9/7/2018 to 21/0/2018– 1^{st} year B.Tech. introduction program	16/09/2019 – World Ozone Day	22/12/2019 – National Mathematics Day	22/03/2019 –World Water Day
15/08/2018 – Independence Day	September – Intramurals	26/01/2019 – Republic Day	23/03/2019 – World Meteorological Day
05/09/2019 – Teachers Day	11/11/2019 – National Education Day	08/03/2019 – International Women's Day	24/03/2019 – Earth Hour
	03/12/2019 – Antipollution Day		In the month of March – Association Days

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science &Engineering (Accredited by NBA)

Class: 4th B.Tech 1st Semester w.e.f. 10-06-2019

Section – A **Class Coordinator :Mr.A. Venkannababu** **Room No: B-303**

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	OT	SA&DP LAB			LUNCH BREAK	CNS	SA&DP	WT
Tue	WT	SA&DP	CNS	MEFA		MC	SPM	CNS
Wed	SA&DP	SPM	WT	MC		CNS	MEFA	LIB/SPO
Thu	OT	WT LAB				MC	SPM	MEFA
Fri	MEFA	MC	SPM	WT		TT		SA&DP
Sat	SPM	MC	SA&DP	CNS		MEFA	WT	TT

Section – B **Class Coordinator: Mrs.D.SasiRekha** **Room No: B-304**

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	WT	SA&DP	SPM	MEFA	LUNCH BREAK	TT LAB		
Tue	OT	SA&DP LAB				WT	SA&DP	SPM
Wed	MC	CNS	MEFA	WT		WT LAB		
Thu	SA&DP	TT LAB				CNS	MEFA	MC
Fri	OT	TT LAB				MC	SPM	CNS
Sat	CNS	SA&DP	SPM	MC		MEFA	WT	LIB/SPO

Section – C **Class Coordinator: Mr.B.Kiran Kumar** **Room No: B-401**

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	WT	MC	CNS	SA&DP	LUNCH BREAK	WT LAB		
Tue	MC	MEFA	CNS	SPM		TT LAB		
Wed	OT	SA&DP LAB				WT	MEFA	SPM
Thu	SPM	CNS	WT	SA&DP		TT LAB		
Fri	CNS	MC	WT	MEFA		SPM	SA&DP	LIB/SPO
Sat	OT	TT LAB				MC	SA&DP	MEFA

Section – D **Class Coordinator: Mr.Y.V.R.P.S.Sastry** **Room No: B-402**

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	SPM	WT LAB			LUNCH BREAK	MEFA	WT	CNS
Tue	SPM	TT LAB				CNS	SA&DP	MEFA
Wed	OT	CNS	SA&DP	WT		SA&DP LAB		
Thu	MC	SA&DP	CNS	SPM		MEFA	MC	LIB/SPO
Fri	SA&DP	SPM	WT	MC		TT LAB		
Sat	OT	MEFA	MC	WT		TT LAB		

Staff Details:

S. No.	*Course Code*	*Course Name*	*Section*			
			A	*B*	*C*	*D*
1.		**CNS**	Mr.N.Praveen Kumar	Mr.B.Kiran Kumar	Mr.B.Kiran Kumar	Mr.N.Praveen Kumar
2.		**SA&DP**	Mr.M.Vamsi Krishna	Ms.D.Sasirekha	Ms.D.Sasirekha	Mr.M.Vamsi Krishna
3.		**WT**	Mr.A.Venkanna Babu	Mr.YVRPS Sastry	Dr.J.Veera Rahgavan	Mr.YVRPS Sastry
4.		**MEFA**	R.V.Raja Sekhar	Dr.K.Rambabu	Dr.K.Rambabu	R.V.Raja Sekhar
5.		**MC**	Mrs.D A S Devi	Mrs.D A S Devi	Ms.P.Sukanya	Ms.P.Sukanya
6.		**SPM**	Dr.V.Venkateswara Rao	Ms.M.RamaRajeswari	Mr.N V Murali krishna Raja	Ms.M.RamaRajeswari
7.		**SA&DP Lab**	Mr.M.Vamsi Krishna, Mrs.D A S Devi	Ms.D.Sasirekha, Ms.P.Sukanya	Ms.D.Sasirekha, Mr.Y. Ravi Raju	Mr.M.Vamsi Krishna, Mrs.D A S Devi
8.		**WT Lab**	Mr.A.Venkanna Babu, Mrs.P.Praneetha	Mr.YVRPS Sastry, Mr.B.Kiran Kumar	Dr.J.Veera Rahgavan, Mr.B.Kiran Kumar	Mr.YVRPS Sastry, Mr.N.Praveen Kumar
9.		**Aptitude**	Mr.J.N.V.Somyajulu	Mr.B.suresh	Mr.J.N.V.Somyajulu	Mr.J.N.V.Somyajulu

Head of the Department

COURSE STRUCTURE

IV B. Tech –I Semester

IV Year –I Semester

S. No.	Subjects	L	T	P	Credits
1	Cryptography and Network Security	4	--	--	3
2	Software Architecture & Design Patterns	4	--	--	3
3	Web Technologies	4	--	--	3
4- HS	Managerial Economics and Financial Analysis	4	--	--	3
5	**Elective-I** i. Big Data Analytics ii. Information Retrieval Systems iii. Mobile Computing	4	--	--	3
6	**Elective-II** i. Cloud Computing ii. Software Project Management iii. Scripting Languages	4	--	--	3
7	Software Architecture& Design Patterns Lab	--	--	3	2
8	Web Technologies Lab	--	--	3	2
	Total Credits				**22**

LESSON PLANS

Cryptography and Network Security

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: IV/I **Section: A,B,C& D**
Name of the Course: Cryptography and Network Security **Course Code:C401**

LESSON PLAN

Course Outcomes (Along with Knowledge Level):
After completion of this course, Student will be able to:

CO	Course Outcome	Knowledge Level
CO1	Discuss the concept of Cryptographic attacks, services and mechanisms.	K2
CO2	Illustrate the various Symmetric key Cryptography algorithms.	K2
CO3	Demonstrate the Number theory and various Asymmetric key Cryptography algorithms.	K3
CO4	Discuss about HASH functions & Digital Signatures to provide secure communication.	K2
CO5	Explain the security at transport layer-TLS and SSL.	K2
CO6	Describe the overview of IP Security and System security.	K2

TEXT BOOKS:
1) Cryptography and Network Security, Behrouz A Forouzan, DebdeepMukhopadhyay, (3e) Mc Graw Hill.
2) Cryptography and Network Security, William Stallings, (6e) Pearson.

REFERENCE BOOKS:

1.Everyday Cryptography, Fundamental Principles & Applications, Keith Martin, Oxford
2. Network Security & Cryptography, Bernard Menezes, Cengage,2010

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	65
2	60	65
3	60	65
4	60	60
5	60	60
6	60	60

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Classify Security Goals	K1	2	Lecture	BB
2		Explain about Security attacks and services	K2	2	Lecture	BB
4		Discuss security mechanisms	K2	1	Lecture	BB
5		Discuss mathematics of cryptography	K2	1	Lecture	BB
6		Discuss Integer Arithmetic	K2	2	Discussion	ICT
7		Discuss Modular Arithmetic	K2	2	Lecture	BB
8		Discuss Matrices and linear congruence	K2	2	Lecture	BB
		TOTAL	**12**			

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Introduction to mathematics of symmetric key cryptography	K1	1	Lecture	BB
2		Explain Algebric structures	K2	2	Lecture	BB
3		Discuss Rings and Fields	K2	1	Lecture	BB
4		Explain GF Fields	K2	1	Discussion	ICT
5		Discuss polynomials and generators	K2	1	Discussion	ICT+BB
6		Explain modern block ciphers	K2	1	Lecture	BB
7		Discuss substitution and transposition	K1	1	Lecture	BB
8		Describe block ciphers as permutations	K2	1	Lecture	BB
9		Discuss S-boxes	K2	1	Lecture	BB
10		Explain modern stream ciphers	K2	1	Lecture	BB
11		Explain DES	K2	2	Lecture	BB
12		Explain AES	K2	1	Lecture	BB
		TOTAL	14			

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Illustrate Prime and Relatively Prime Numbers	K2	2	Lecture	BB
2		Explain Modular Arithmetic	K2	1	Lecture	BB
3		Illustrate about Fermat's and Euler's Theorems	K3	2	Lecture	BB
4		Explain The Chinese Remainder theorem	K2	1	Discussion With Lecture	ICT
5		Discuss quadratic congruence	K3	1	Lecture	BB
6		Describe Exponentiation and logarithm	K2	1	ICT	BB
7		Illustrate RSA Algorithms	K3	1	Lecture	BB+ICT
8		IllustrateDiffie Hellman KeyExchange	K3	1	Lecture	BB
9		Explain Elgamal encryption & decryption	K2	1	Lecture	BB
10		Discuss Elliptic Curve Cryptography	K2	1	Lecture	BB
		TOTAL		**12**		

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Discuss about Message Integrity functions	K2	1	Lecture	BB
2		Explain about Message Authentication Functions	K2	1	Lecture	BB
3		Discuss about Application of Cryptographic hash Functions	K2	1	Lecture	BB
4		Explain Secure Hash Algorithm	K2	2	Lecture With Discussion	BB+ICT
5		Explain about Message Authentication Functions	K2	1	Lecture	BB
6		Distinguish HMAC & CMAC	K2	2	Discussion	BB
7		Explain about Digital Signatures schemes	K2	2	Lecture with discussion	BB+ICT
8		Explain attacks on digital signature	K2	1	Lecture	BB
9		Discuss variations and applications of digital signatures	K2	1	Lecture	BB+ICT
		TOTAL		**12**		

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Explain Electronic Mail Security	K2	1	Lecture	BB
2		Discuss pretty good privacy(PGP)	K2	1	Lecture	BB
3		Discuss about S/MIME	K2	2	Lecture With Discussion	BB+ICT
4		Explain about Secure Socket Layer (SSL)	K2	1	Lecture	BB
5		Explain Transport Layer Security(TLS)	K2	1	Discussion	BB
		TOTAL		**06**		

CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Outline IP Security Overview	K1	1	Lecture	BB
2		Explain IP Security Architecture	K2	1	Lecture	BB
3		Explain Authentication Header	K2	1	Lecture	BB
4		Discuss about Encapsulating Security Payload	K2	1	Lecture	BB+ICT
5		Explain Internet key exchange mechanism	K2	1	Lecture	BB
6		Discuss about system security	K2	2	Lecture	BB
7		**TOTAL**	**07**			

Total No. of Classes: 63

SOFTWARE ARCHITECTURE & DESIGN PATTERNS

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: IV/I **Section: A,B,C& D**
Name of the Course: SOFTWARE ARCHITECTURE & DESIGN PATTERNS
Course Code: 402

LESSON PLAN

Course Outcomes (Along with Knowledge Level):
After completion of this course, Student will be able to:

CO1. Define various Architecture reference models and Quality Attributes. (K1)

CO2. Explain the mechanism of Evaluating Architecture. (K2)

CO3. Apply various principles in selecting and using Patterns. (K3)

CO4. Construct Structural Patterns for a given Scenario. (K3)

CO5. Construct Behavioral Patterns for a given Scenario. (K3)

CO6. Develop various Case Studies in understanding Software Architectural issues. (K3)

Text Books suggested:

T1. Software Architecture in Practice, Second Edition, Len Bass, Paul Clements & Rick Kazman, Pearson Education, 2003.

T2. Design Patterns, Erich Gamma, Pearson Education, 1995.

Reference Books suggested:

R1. Beyond Software Architecture, Luke Hohmann, Addison Wiley, 2003.

R2. Software Architecture, David M.Dikel, David Kane and James R.Wilson, Prentice Hall PTR, 2001.

R3. Software Design, David Budgen, Second Edition, Pearson Education, 2003.

R4. Head First Design Patterns, Eric Freeman & Elisabeth Freeman, O'REILLY, 2007.

R5. Design Patterns in Java, Steven John Metsker & William C. Wake, Pearson Education, 2006.

R6. J2EE Patterns, Deepak Alur, John Crupi & Dan Malks, Pearson Education, 2003.

R7. Design Patterns in C#, Steven John Metsker, Pearson Education, 2004.

R8. Pattern Oriented Software Architecture, F. Buschmann & Others, John Wiley & Sons.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
402.1	60	60
402.2	60	60
402.3	65	65
402.4	65	65
402.5	65	65
402.6	60	60

Unit-1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Introduction to OBE, Dissemination of Vision, Mission of the Dept.and PEOs,POs & PSOs of the Programme.		1	Lecture	BB
2		Describe the Architecture Business Cycle.	K1	1	Lecture	BB
3		Define Software Architecture.	K1	1	Lecture	BB
4		Outline various Architectural Patterns	K1	1	Lecture	BB+ICT
5		State Reference Models and Architectures	K1	1	Lecture	BB
6		Outline architectural Structures and Views.	K1	1	Lecture	BB+ICT
7		Define Quality Attributes, Describe the steps of Achieving them	K1	1	Lecture with discussion	BB+ICT
8		Describe Architectural Styles and Patterns.	K1	2	Lecture with discussion	BB+ICT
9		State the steps of Designing, Documenting and Reconstructing the Software Architecture.	K1	2	Lecture with discussion	BB

Unit- 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Explain the importance of Software Evaluation.	K2	1	Lecture	BB
2		Infer the role of Architecture Design in Decision Making	K2	1	Lecture with Discussion	BB
3		Describe the Models – ATAM, CBAM	K1	2	Lecture with Discussion	BB+ICT
4		Define Software Product Lines	K1	1	Lecture	BB
5		Interpret the steps of Building Systems from off the shelf components.	K2	2	Lecture with Discussion	BB+ICT
6		Infer the future of Software Architecture	K2	1	Lecture with Discussion	BB+ICT

Unit-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Define Pattern Descriptions	K1	1	Lecture	BB
2		Describe the steps of Organizing Catalogues and their role in solving Design Problems	K2	2	Lecture with Discussion	BB + ICT
3		Apply various principles in Selecting and Using Patterns	K3	2	Lecture with Discussion	BB + ICT
4		Construct an Abstract factory Pattern for a given Scenario	K3	1	Lecture with Discussion	BB+ ICT

5		Describe the Builder Pattern using an example.	K2	1	Lecture with Discussion	BB+ ICT
6		Describe the Factory Method Pattern using an example.	K2	1	Lecture with Discussion	BB+ ICT
7		Describe the Prototype Pattern and its consequences.	K2	1	Lecture with Discussion	BB+ ICT
8		Explain the Singleton Pattern using an example.	K2	1	Lecture with Discussion	BB+ ICT

Unit- 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Describe Adapter Structural Pattern.	K1	1	Lecture	BB
2		Explain Bridge Structural Pattern.	K2	1	Lecture with Discussion	BB+ ICT
3		Construct Composite Structural Pattern for a given scenario.	K3	1	Lecture with Discussion	BB+ ICT
4		Construct Decorator Pattern for a given scenario.	K3	1	Lecture with Discussion	BB+ ICT
5		Explain the Façade Structural Pattern.	K2	1	Lecture with Discussion	BB+ ICT
6		State and explain the Flyweight Structural Pattern.	K2	2	Lecture with Discussion	BB+ ICT
7		Describe PROXY Structural Pattern.	K1	1	Lecture	BB

Unit-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	State and explain the Chain of Responsibility Behavioural Pattern.	K1	1	Lecture	BB
2		Describe the Command Behavioural Pattern.	K2	1	Lecture with Discussion	BB + ICT
3		Develop Interpreter Behavioural Pattern for a given example.	K3	1	Lecture with Discussion	BB + ICT
4		Construct Iterator Behavioural Pattern for a given example.	K3	1	Lecture with Discussion	BB + ICT
5		Construct Mediator Behavioural Pattern for a given example.	K3	1	Lecture with Discussion	BB + ICT
6		Explain the Memento Behavioural Pattern.	K2	1	Lecture with Discussion	BB + ICT
7		Describe Observer, State and Strategy Behavioural Patterns in detail.	K1	3	Lecture with Discussion	BB + ICT
8		State and explain Template Method and Visitor Behavioural Pattern.	K1	2	Lecture	BB

Unit-6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Examine the A-7E Case Study in utilizing architectural structures.	K3	1	Lecture with practical	BB + ICT
2		Assess the World Wide Web Case Study in understanding Interoperability.	K3	1	Lecture with practical	BB + ICT
3		Explain the design steps of Air Traffic Control in achieving high availability.	K2	1	Lecture with Discussion	BB + ICT
4		Describe the design steps of Celsius Tech in dealing product line development.	K1	1	Lecture	BB
5		Define the design problems of a Document Editor	K1	1	Lecture	BB
6		Explain the Document Structure of a Document Editor	K2	1	Lecture with Discussion	BB
7		Explain the Formatting steps of a Document Editor	K2	1	Lecture with Discussion	BB + ICT
8		State the steps of Embellishing the User Interface.	K1	1	Lecture	BB
9		Demonstrate the Supporting steps of Multiple Look-and-Feel Standards, Multiple Window Systems.	K3	2	lecture with Practical	BB + ICT
10		Describe the functionalities – User Operations, Spelling Checking, Hyphenation	K1	2	Lecture	BB

Total No. of Classes: 60

Web Technologies

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: IV/I **Section: A,B,C& D**
Name of the Course: Web Technologies **Course Code:C-403**

LESSON PLAN

Course Outcomes (Along with Knowledge Level):
After completion of this course, Student will be able to:

CO Number	Course Outcome	Knowledge-Level
403.1	Demonstrate HTML tags, Cascading style sheets.	K3
403.2	Demonstrate Java script,DHTML.	K3
403.3	Prepare XML Documents.	K3
403.4	Prepare forms and Database using PHP.	K3
403.5	Apply PERL to Retrieve documents from the web.	K3
403.6	Use Ruby for Practical Web Applications.	K3

Text Books/ Reference Books suggested:

Text Books:

1. Programming the World Wide Web, Robet W Sebesta, 7ed, Pearson.

2. Web Technologies, Uttam K Roy, Oxford

3. The Web Warrior Guide to Web Programming, Bai, Ekedahl, Farrelll, Gosselin, Zak, Karparhi, Maclntyre, Morrissey, Cengage.

Reference Books:

1. Ruby on Rails Up and Running, Lightning fast Web development, Bruce Tate, Curt Hibbs, Oreilly (2006).

2. Programming Perl, 4ed, Tom Christiansen, Jonathan Orwant, Oreilly (2012)

3. Web Technologies, HTML< JavaScript, PHP, Java, JSP, XML and AJAX, Black book, Dream Tech.

4. An Introduction to Web Design, Programming, Paul S Wang, Sanda S Katila, Cengage.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	60
2	60	60
3	60	60
4	60	60
5	60	60
6	60	60

SNo	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	**CO1**	Describe HTML tags	K1	3	Lecture	BB
		Explain Lists, Tables, Images	K2	3	Lecture	BB+ICT
		Explain Forms, Frames	K2	2	Lecture	BB+ICT
		Explain HTML5	K2	2	Lecture	BB+ICT
		Define Cascading style sheets	K1	3	Lecture	BB
2	**CO2**	Explain Objects, Primitives Operations and Expressions of JavaScript	K2	3	Lecture	BB
		Explain Output and Keyboard Input, Control Statements	K2	2	Lecture	BB
		Explain Object Creation and Modification, Arrays, Functions	K2	2	Lecture	BB
		Explain Constructors, Pattern Matching using Regular Expressions	K2	2	Lecture	BB
		Apply Positioning Moving and Changing Elements with DHTML	K3	2	Lecture	BB+ICT
3	**CO3**	Explain XML, Document type Definition, XML schemas	K2	3	Lecture	BB
		Demonstrate Document object model, XSLT, DOM and SAX.	K3	3	Lecture with Discussion	BB
		Explain AJAX	K2	1	Lecture	BB+ICT

		Apply Integrating PHP and AJAX	K3	2	Lecture with Discussion	BB+ICT
4	**CO4**	Explain Creating PHP script,	K2	2	Lecture	BB
		Demonstrate Running PHP script. Working with variables and constants: Using variables, Using constants, Data types, Operators	K3	3	Lecture with Discussion	BB+ICT
		Classify Conditional statements, Control statements	K2	2	Lecture	BB
		Describe Arrays, functions	K2	2	Lecture	BB
		Apply forms and Databases such as mySql, Oracle, SQL Sever.	K3	2	Lecture	BB+ICT
5	**CO5**	Explain PERL Operators and IF Statements	K2	3	Lecture	BB+ICT
		Describe Program Design and Control Structures	K2	2	Lecture	BB
		Explain Arrays,Hashes and File Handling	K2	2	Lecture	BB
		Describe Regular Expressions,Subroutines	K2	1	Lecture	BB+ICT
		Demonstrate Retrieving Documents from the web with PERL	K3	1	Lecture	BB+ICT
6	**CO6**	Explain Ruby, variables, types, simple I/O	K2	2	Lecture	BB
		Demonstrate Control, Arrays, Hashes, Methods, Classes	K3	2	Lecture	BB+ICT
		Iterators, Pattern Matching	K3	1	Lecture	BB
		Explain Overview of Rails	K2	2	Lecture	BB+ICT

TOTAL: 60

Managerial Economics and Financial Analysis

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: IV/I **Section: A,B,C& D**
Name of the Course: Managerial Economics and Financial Analysis
Course Code: C404

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):
After completion of this course, the students will be able to:

CO	Course Outcome	Knowledge Level
404.1	Estimating the demand for a product and the relationship between price and demand.	(K2)
404.2	Interpret production concept, least cost combinations and various costs concepts in decision making.	(K3)
404.3	Describe the price –output determination under various market structures.	(K2)
404.4	Classify various forms of business organization.	(K2)
404.5	Prepare the financial statements to gains knowledge to analyze them.	(K3)
404.6	Assess various investment project proposals with the help of Capital Budgeting techniques for decision making.	(K3)

Text Books/ Reference Books suggested:

1. Aryasri: Managerial Economics and Financial Analysis, 2/e, TMH, 2005. 2.
2. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2003.
3. S A Siddhiqui &AS Siddhiqui , Managerial Economics and Financial Analysis, New Age international publishers 2013.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	60
2	60	60
3	60	60
4	60	60
5	60	60
6	60	60

S. no	Course Outcome	Contents	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids	Remarks
	Estimate the demand for a product and the relationship between price and demand.	**UNIT–Introduction to managerial economics**		10			
1		Define managerial economics	K1	1	Lecture Discuss	Black board	
2		Describe ME with other disciplines	K1	1	Lecture	Black board	
3		Explain Nature and scope of managerial economics	K2	1	Lecture	Black board	
4		Define Demand	K1	1	Lecture Discuss	Black board	
5		Describe law of demand	K2	1	Lecture	Black board	
6		Explain Elasticity of demand	K2	1	Lecture	Black board	
7		Find the of elasticity of demand	K2	2	Lecture	Black board	
8		Explain Demand forecasting, methods.	K2	2	Lecture Discuss	Black board	
		UNIT–II: Production analysis an d cost analysis		08			
1	Interpret production concept, least cost combinations and various costs concepts in decision making.	State Production function	K1	1	Lecture	Black board	
2		State Iso-cost	K1	1	Lecture	Black board	
3		State Iso-quants	K1	1	Lecture	Black board	
4		Explain Cob-Douglas production function	K2	1	Lecture Discuss	Black board	
5		Describe economies of scale	K2	1	Lecture	Black board	
		Enumerate various cost concepts	K1	1	Lecture	Black board	
		Solve problems in break even analysis	K3	2	Lecture	Black board	
		UNIT–III: market structures		10			
1	Describe the price – output determinat	Describe Different types of market structures	K1	3	Lecture	Black board	
2		Explain Price-output determination	K2	3	Lecture	Black board	

3		Explain Pricing methods	K2	2	Lecture	Black board	
4		Describe Theories of firm	K2	2	Lecture	Black board	
		UNIT–IV: Business organization		06			
1	Classify various forms of business	State Features, merits ,demerits of Different forms of Organizations	K1	4	Lecture	Black board	
2		Differentiate various forms of business organizations	K2	2	Lecture	Black board	
		Unit–V: financial accounting		12			
1	Prepare the financial statements to gains knowledge to analyze them.	Describe Accounting Cycle	K1	1	Lecture	Black board	
		Explain Double Entry System	K2	1	Lecture	Black board	
		Prepare final Accounts	K3	4	Lecture	Black board	
2		Examine financial statements by using funds flow, Ratio analysis	K3	6	Lecture	Black board	
		UNIT–VI: capital budgeting		08			
1	Assess various investment project proposals with the help of Capital	Define Capital	K1	1	Lecture	Black board	
2		Enumerate types of capital	K1	1	Lecture	Black board	
3		Explain capital budgeting, Process	K2	2	Lecture	Black board	
5		Apply capital budgeting techniques	K3	4	Lecture	Black board	

Total No. of Classes: 54

Mobile Computing

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: IV/I **Section: A,B,C& D**
Name of the Course: Mobile Computing **Course Code: C-405**

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, students will be able to:

CO.1 Describe mobile computing Paradigm and its applications,limitations. [K2]
CO.2 Discuss wireless networks. [K2]
CO.3 Demonstrate various layers of mobile networks. [K3]
CO.4 Develop mobile transport layer and database issues. [K3]
CO.5 Illustrate different data delivery methods and synchronization protocols . [K3]
CO.6 Demonstrate mobile adhoc networks and related concepts. [K3]

TEXT BOOKS:

1. Jochen Schiller, “Mobile Communications”, Addison-Wesley, Second Edition, 2009.
2. Raj Kamal, “Mobile Computing”, Oxford University Press, 2007, ISBN: 0195686772

REFERENCE BOOKS:

1. ASOKE K TALUKDER, HASAN AHMED, ROOPA R YAVAGAL, “Mobile Computing, Technology Applications and Service Creation” Second Edition, Mc Graw Hill.
2. UWE Hansmann, LotherMerk, Martin S. Nocklous, Thomas Stober, “Principles of Mobile Computing,” Second Edition, Springer.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	70	70
2	65	60
3	65	60
4	60	60
5	60	60
6	60	60

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission of the Dept.and PEOs,Pos,& PSOs of the Programme		1	Lecture	BB
2		Describe Mobile Communications	K1	1	Lecture	BB
3		Describe Mobile Computing – Paradigm, Promises/Novel Applications and Impediments and Architecture .	K1	1	Lecture	BB
4		Discuss Mobile and Handheld Devices, Limitations of Mobile and Handheld Devices.	K2	1	Lecture With Discussion	BB
5		Discuss GSM – Services, System Architecture, Radio Interfaces, Protocols, Localization, Calling, Handover, Security, New Data Services, GPRS.	K2	2	Lecture	BB

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Describe (Wireless) Medium Access Control (MAC), Wireless.	K1	1	Lecture	BB
2		Explain Motivation for a specialized MAC.	K2	2	Lecture with Discussion	BB
3		Describe SDMA, FDMA.	K1	2	Lecture with Discussion	BB
4		Discuss TDMA.	K2	1	Lecture	BB+ICT
5		Explain CDMA.	K2	2	Lecture with Discussion	BB+ICT
6		Discuss Wireless LAN.	K2	2	Lecture with Discussion and in class Assignment	BB+CRE

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Describe Mobile Network Layer: IP and Mobile IP.	K1	1	Lecture	BB
2		Illustrate Network Layers, Packet Delivery and Handover Management.	K3	3	Lecture	BB
3		Explain Location Management, Registration.	K2	1	Lecture with Discussion	BB
4		Discuss Tunneling and Encapsulation.	K2	2	Lecture	BB+ICT
5		Demonstrate Route Optimization.	K3	2	Lecture with Discussion	BB+ICT
6		Demonstrate DHCP.	K3	2	Lecture with Discussion	BB+ICT

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Describe Mobile Transport Layer.	K1	1	Lecture	BB
2		Demonstrate Conventional TCP/IP Protocols.	K3	2	Lecture with Discussion	BB
3		Discuss Indirect TCP.	K2	3	Lecture with Discussion	BB
4		Demonstrate Snooping TCP.	K3	2	Lecture with Discussion	BB+ICT
5		Demonstrate Mobile TCP, Other Transport Layer Protocols for Mobile Networks.	K3	3	Lecture with Discussion	BB+ICT
6		Explain Database Issues Database Hoarding & Caching Techniques.	K2	1	Lecture with Discussion	BB+ICT

7		Discuss Client-Server Computing & Adaptation.	K2	1	Lecture with Discussion	BB+ICT
8		Discuss Transactional Models, Query processing, Data Recovery Process &QoS Issues.	K2	1	Lecture with Discussion	BB+ICT

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Describe Data Dissemination and Synchronization.	K1	2	Lecture	BB
2		Discuss Classification of Data Delivery Mechanisms.	K2	2	Lecture with Discussion	BB
3		Explain Communications Asymmetry, Data Dissemination.	K2	2	Lecture	BB
4		Compute Broadcast Models, Selective Tuning and Indexing Methods.	K3	3	Lecture with Discussion	BB
5		Discuss Data Synchronization – Introduction, Software, and Protocols.	K2	1	Lecture with Discussion	BB

CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Describe Mobile Ad hoc Networks (MANETs) Introduction.	K1	1	Lecture	BB
2		Discuss Applications & Challenges of a MANET, Routing.	K2	2	Lecture with Discussion	BB
3		Explain Classification of Routing Algorithms, Algorithms such as DSR, AODV, DSDV, etc. ,	K2	2	Lecture with Discussion	BB

		Mobile Agents, Service Discovery.				
4		Show Protocols and Platforms for Mobile Computing :WAP.	K3	1	Lecture with Discussion	BB
5		Show Bluetooth, XML, J2ME, Java Card, PalmOS, Windows CE.	K3	2	Lecture with Discussion	BB
6		Explain SymbianOS, Linux for Mobile Devices, Android.	K2	1	Lecture with Discussion	BB

Total No. of Classes: 60

Software Project Management

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: IV/I **Section: A,B,C& D**
Name of the Course: Software Project Management **Course Code: C-406**

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):
After completion of this course, the students will be able to:

CO1. Define Software Project Management Terminology and Methodology. (K1)

CO2. Describe various Software Lifecycle Models, Process Artifacts and Workflows. (K2)

CO3. Explain various Effort Estimation Techniques for Project Planning. (K2)

CO4. Demonstrate Risk Management Concepts. (K3)

CO5. Develop Project Status Reports for tracking and controlling Software Deliverables. (K3)

CO6. Describe Software Quality Metrics. (K2)

Text Books suggested:

T1. Software Project Management, Bob Hughes & Mike Cotterell, TMH

T2. Software Project Management, Walker Royce, Pearson Education, 2005.

T3. Software Project Management in Practice, Pankaj Jalote, Pearson

Reference Books suggested:

R1. Software Project Management, Joel Henry, Pearson Education.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
406.1	65	60
406.2	65	60
406.3	60	60
406.4	60	60
406.5	60	60
406.6	65	60
406	63	60

Unit-1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1		Dissemination of Vision, Mission of the Dept and PEOs, Pos,& PSOs of the Programme			Lecture	BB
2	CO1	Define Software Project Management Terminology.	K1	1	Lecture	BB
3		Describe software project management activities.	K1	1	Lecture	BB
4		Describe various challenges in software Projects	K1	1	Lecture	BB
5		Identify types of stake holders, objectives and goals in software project management.	K1	2	Lecture	BB
6		Describe Stepwise project planning and project scope.	K1	1	Lecture	BB
7		Identify Project products and Deliverables.	K1	1	Lecture	BB
8		Outline Effort Estimation and Infrastructure.	K1	1	Lecture	BB+ICT

Unit- 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Outline various Life Cycle Models.	K1	1	Lecture	BB
2		Classify technologies: Process Models	K2	2	Lecture	BB
3		Describe Software Prototyping.	K2	1	Lecture	BB
4		Explain Iterative and Incremental Process Framework.	K2	1	Lecture	BB
5		Classify Project Life Cycle Phases.	K2	1	Lecture	BB+ICT
6		Explain various Artifacts of Software Process.	K2	2	Lecture	BB
7		Explain Process Workflows.	K2	2	Lecture	BB

Unit-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Describe Software Effort Estimation Techniques.	K1	1	lecture	BB
2		Discuss Function Point Analysis.	K2	1	lecture with Discussion	BB
3		Explain SLOC: Software Metrics and Measurements.	K2	2	lecture	BB + ICT
4		Describe COCOMO: A Parametric Model	K2	2	lecture	BB + ICT
5		Discuss Use-Case based Estimation Techniques.	K2	1	lecture with Discussion	BB
6		Explain various Activity Identification Approaches: Sequencing and Scheduling Activities.	K2	2	lecture	BB
7		Discuss Network Planning Models in Project Scheduling: Critical Path Analysis.	K2	2	lecture with Discussion	BB

Unit- 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**CO4**	Describe various Risk Management Categories.	K1	2	Lecture	BB
2		Discuss concepts of Risk Identification, Assessment, Planning and Management.	K2	2	Lecture with discussion	BB+ICT
3		Demonstrate PERT Technique.	K3	1	Lecture	BB
4		Explain Monte Carlo Method for project estimation.	K2	2	Lecture	BB

Unit-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**CO5**	Describe the concept of Project Monitoring and Control.	K1	1	lecture	BB
2		Explain Progress Monitoring, and Cost Monitoring in Project Control.	K2	2	lecture	BB
3		Explain Earned Value Analysis in Cost Monitoring.	K2	2	lecture	BB + ICT
4		Discuss various Defects and Issues in Project Monitoring and Control.	K2	1	Lecture	BB + ICT
5		Develop Project Status Reports with Sample Case Study.	K3	1	lecture	BB
6		Discuss various types of resources and resource requirements in Software Project Management.	K2	2	lecture with discussion	BB

7		Explain the concept of Resource Allocation and Scheduling.	K2	1	Lecture with practical	BB

Unit-6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**CO6**	Define Software Quality and Quality Factors.	K1	1	Lecture	BB + ICT
2		Explain Software Quality Planning.	K2	1	Lecture	BB + ICT
3		Outline various Quality Measures in Software Quality Management.	K1	2	Lecture	BB + ICT
4		Discuss Quantitative Approaches to Quality Management.	K2	2	Lecture with Discussion	BB + ICT
5		Describe Quantitative Quality Management Planning to achieve Quality Goals.	K2	1	Lecture	BB + ICT
6		Explain the concepts of product Quality and Process Quality.	K2	1	Lecture	BB + ICT
7		Describe Statistical Process Control Capability Maturity Model.	K2	2	Lecture with Discussion	BB + ICT
8		Discuss various Techniques to Enhance Software Quality.	K2	1	Lecture with Discussion	BB + ICT

Total No. of Classes: 57

Software Architecture and Design Patterns Lab

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: IV/I **Section: A,B,C& D**
Name of the Course: Software Architecture and Design Patterns Lab
Course Code: 407

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):
After completion of this course, the students will be able to:

407.1	Develop the Use Case and Logical views for a given scenario.(K3)
407.2	Demonstrate the integration of Architectural and Design Patterns in the logical view. (K3)
407.3	Construct the Implementation, Process, Deployment Views for Weather Mapping System.(K3)
407.4	Develop the generation and implementation strategies for Component and In Process Communication.(K3)
407.5	Illustrate various Structural design patterns like Adapter, Bridge, Proxy etc.(K3)
407.6	Demonstrate various Behavioral design patterns like Strategy, Iterator, Visitor etc.(K3)

WEB TECHNOLOGIES LAB

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: IV/I **Section: A,B,C& D**
Name of the Course: WEB TECHNOLOGIES LAB **Code: C-408**

LESSON PLAN

Course Outcomes
Upon Completion of the course, student will be able to:

408.1	Prepare dynamic websites with good aesthetic sense of designing and latest technical know-how's.(K3)
408.2	Prepare XML Documents . (K3)
408.3	Use Ruby for Practical Web Applications. (K3)
408.4	Develop Interface with CGI-PERL. (K3)
408.5	Prepare forms and Databases(MySQL) using PHP. (K3)